

1. List 5 common hardware issue you might encounter in a desktop PC and write the typical symptom for each.

→ * Failing power supply (PSU) :-

Typical symptom :- PC will not turn on no lights, power LEDs or spinning fans when you press the power button.

* Faulty RAM :-

Typical symptom :- PC turn on but show blank screen, continuous beep codes, random Blue Screen of death (BSOD) crashes.

* CPU overheating :-

Typical symptom :- fans spin at maximum speed and the PC suddenly turn off after running for few minutes.

* Failing Hard Drive / SSD :-

Typical symptom :- system freezes frequently, files open extremely slowly or a "No Bootable device Found" error message appears.

* Faulty graphics card (GPU) :-

Typical symptom :- screen display strange colors, lines or visual glitches or monitor signal cuts out during gaming.

2. Given this scenario: A laptop is not charging even when plugged it in. write down a step by step troubleshooting plan using only basic tools and explain what you need check at each step.

→ 1. check the wall outlet & charger LED :-

plug lamp/bulb into same outlet make it on. or not.

2. Inspect the cable & plug :-

Look for the loose or broken pin inside charging tip

3. Inspect the Charging Port :-

check with flashlight in charging port in laptop. look for dust buildup, debris or wobbly connector pin inside.

4. Test power brick output multimeter :-

Touch multimeter probes to charger's barrel plug (USB C pin) verify if the plug voltage matches the rated sticker label.

5. perform Hard power Reset :-

Unplug charger, remove battery and hold the laptop button down for 30 seconds to drain residual static charge. Reconnect power & check. test.

6. Inspect the Internal Battery :-

open back panel & examine battery. Look for physical swelling, bulging cases or loose ribbon connectors cable wired to the motherboard.

3. Use multimeter to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range.

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* multimeter setting :-

DC voltage mode (set dial to 20V DC)

* Test method :-

probes placed on the outer power pins^{pins} (pin 1 is +5V, pin 4 ground) of a powered USB Type-A port.

* measured Reading :- 5.08V

Verdict :- Normal/Healthy. The reading of 5.08V is well within the expected standard USB range of 4.75 to 5.25V.

4. A PC is randomly shutting down, use a multimeter to test power supply's 12V rail (yellow wire) from disconnect ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V what hardware issue could this indicate?

Steps :-

1. Safety :- Unplug power supply from the wall and disconnect 24 pin ATX connector from the motherboard.

2. Paperclip Jump :- Insert a paperclip into the green wire (PS-ON) pin and any Black wire (ground) pin to manually jump start the power supply. Plug the power cord back in and flip its switch on.

3. multimeter Setup :-

Turn the multimeter dial to DC voltage

4. probe :-

Insert the Red probe into a slot with a yellow wire (+12V) and the Black probe into a slot with a Black wire (ground).

* measured Result & diagnosis :-

* simulated Reading : 10.8V

* Hardware Issue :- A voltage below 11.5V indicates a failing power supply unit (PSU). The 12V rail supplies critical power to high-draw components like the CPU and graphic card. Under voltage forces the PC to crash or suddenly shut down under load to protect internal hardware.